COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	07-29-2026	**DATE SUBMITTED:**	07-29-2026
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Seatwork 4.1 **Array-Based vs. STL Stack Implementations**

1. PROCEDURE OUTPUT

What procedures did you perform?

ILOA

```cpp
#include <iostream>
#include <stack>

int main(){

    std::stack<int> stack1;
    std::stack<int> stack2;
    std::cout<<"Testing the stack STL"<<std::endl;

    //isEmpty
    std::cout<<"is the stack empty? "<< stack1.empty()<<std::endl;
    //push
    stack1.push(10);
    std::cout<< "The top of the stack is " << stack1.top()<<std::endl;

    stack1.push(9);
    std::cout<< "The top of the stack is " << stack1.top()<<std::endl;

    stack1.push(8);
    std::cout<< "The top of the stack is " << stack1.top()<<std::endl;

    stack1.push(7);
    std::cout<< "The top of the stack is " << stack1.top()<<std::endl;

    stack1.pop();
    std::cout<<"The top of the stack is "<< stack1.top() <<std::endl;
```

```
    std::cout<<"is the stack empty? "<<stack1.empty()<<std::endl;

    std::cout<<"the size of the stack is: " << stack1.size()<<std::endl;



    stack1.emplace(6);
    std::cout<<"the top of the stack is " <<stack1.top() <<std::endl;

    stack1.emplace(5);
    std::cout<<"the top of the stack is " <<stack1.top() <<std::endl;

    stack1.emplace(4);
    std::cout<<"the top of the stack is " <<stack1.top() <<std::endl;

    stack1.emplace(3);
    std::cout<<"the top of the stack is " <<stack1.top() <<std::endl;


    stack1.swap(stack2);
    std::cout << "After swap, stack1 size is: " << stack1.size() <<
std::endl;




    return 0;









}
```

```
Testing the stack STL
is the stack empty? 1
The top of the stack is 10
The top of the stack is 9
The top of the stack is 8
The top of the stack is 7
The top of the stack is 8
is the stack empty? 0
the size of the stack is: 3
the top of the stack is 6
the top of the stack is 5
the top of the stack is 4
the top of the stack is 3
After swap, stack1 size is: 0
PS C:\Users\TIPQC\OneDrive\Documents\CPE21S2 - Milan>
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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